

SHUBHAM LOKARE

Generative AI Engineer | AI Engineer | Data Scientist

91+ 9130884843 | shubhamlokare4843@gmail.com | <https://github.com/shubhaml4843>

🔗 [linkedin.com/in/shubhamlokare-aiengineer/](https://www.linkedin.com/in/shubhamlokare-aiengineer/) | 🌐 shubhaml4843.github.io/Portfolio/

Profile Summary

AWS Machine Learning Certified AI Engineer with 1+ years of experience in Generative AI, LLMs, RAG, Diffusion Models, and NLP. Skilled in building agentic AI workflows using LangChain, LangGraph, and MCP, and experienced in fine-tuning with LoRA, Q-LoRA, and RLHF. Proven ability to develop and deploy ML/DL models end-to-end, delivering impactful real-world AI solutions.

Technical Skills

- **Generative AI / AI:** Chatbots, LLMs, RLHF, Diffusion Models, GANs, VAEs, RAG Pipeline, Agentic AI, NLP, Transformers, Embeddings, Multimodal AI, Vector Databases (Pinecone, FAISS, Weaviate).
- **Prompt Engineering & Fine-Tuning:** LoRA, Q-LoRA, Advanced Prompting, Few-shot Learning, ReAct, Chain-of-Thought (CoT), Tree-of-Thought (ToT), Evaluation Metrics (BLEU, ROUGE).
- **ML/DL:** Regression, Random Forest, XGBoost, LightGBM, SVM, ANN, CNN, RNN, LSTM, Computer Vision, YOLO, Clustering, Dimensionality Reduction, Ensemble Learning.
- **Frameworks & Tools:** LangChain, LangGraph, AutoGen, Hugging Face, MCP, LLM Orchestration, Multimodal Orchestrators (LlamaIndex, Haystack), OpenAI API, Ollama API, Cohere API.
- **Languages & Libraries:** Python, SQL, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, BERT, OpenCV, Streamlit.
- **Platforms:** AWS (EC2, S3, Lambda, SageMaker), Docker, Git, Nginx.

Professional Experience

AI Engineer

ImmersiveVision Technology Pvt.Ltd | Dec 2024 – Present

- **Led end-to-end AI projects for Cadaviz – a 3D Virtual Human Dissection Table**, including design, development, and **deployment of RAG pipelines**, chatbots (Hugging Face, Cohere, Llama API), **MCP workflows**, and **Whisper-based voice solutions** — **improving query response accuracy by 28%**.
- Built and integrated **chatbots for feedback and concept explanation**, enabling medical students to interact with the dissection table through Q&A, search, and **voice commands** — **increasing user adoption by 35%**.
- Developed and **fine-tuned Generative AI, NLP, and computer vision** models for **object detection, radiology/anatomy analysis, multimodal processing** — **boosting diagnostic accuracy by 18%**.
- Conducted data analysis, model optimization, and **accuracy improvements**, delivering optimized solutions and **real-time dashboards (Power BI)** — **cutting report generation time by 60%**.
- Managed **full-stack AI deployment using AWS, Docker, and FastAPI**, while mentoring junior team members — **reducing deployment cycle time by 70% and training 4+ engineers in AI workflows**.

Data Scientist Intern

AISPR | July 2024 – Dec 2024

- Conducted comprehensive **EDA, data preprocessing, and feature engineering**, optimizing datasets to **improve ML/DL model performance by 10–15%**.
- Developed and **fine-tuned ML models (Random Forest, XGBoost, LightGBM, SVM)** and DL models (**ANN, CNN, RNN, LSTM, YOLO**), achieving high accuracy and optimized predictions.
- Implemented **time-series forecasting predictive analytics solutions**, **delivering production-ready results across 3 projects**.

Selected Projects

DataSynapse AI – Multi-Agent AI Platform with RLHF, RAG & MCP

[GitHub](#) | [Demo](#)

Problem Solved: Fragmented AI workflows force engineers to spend **60–70%** of their time manually coordinating tasks **across ML/DL, NLP, code execution, and visualization, causing inefficiency and delays.**

- Designed and implemented a **production-ready 10-agent AI system** with Core Orchestrator & MCP Dispatcher, automating workflows across **Full EDA, data processing, ML/DL, NLP and visualization.**
- Leveraged **LangGraph** for multi-agent coordination, achieving **~90% automation**, optimizing **routing, task execution, and parallel processing** for AI engineers and data scientists reducing manual engineering effort by **65%.**
- Integrated **RLHF (Reinforcement Learning from Human Feedback)** to continuously improve agent performance and workflow efficiency.
- Built **RAG pipelines** using external knowledge sources (**Wikipedia, APIs, web search**) with **vector memory** for **context-aware, real-time responses.**
- Applied **advanced prompting techniques (ReAct, CoT, ToT)** and implemented **self-attention mechanisms** to improve query understanding, reasoning by **22%.**
- Developed **FastAPI web interface** and CLI runner, enabling scalable, high-performance AI system engineering with memory management and robust logging — cutting experiment cycle time by **40%**
- Demonstrated expertise in **AI workflow automation, LLM fine-tuning (LoRA, Q-LoRA), and enterprise-grade system optimization** in complex multi-agent orchestration scenarios.

Medical ChatBot – AI-Powered Clinical Assistant with RAG & NLP

[GitHub](#) | [Demo](#)

Problem Solved: Manual review of clinical reports is **slow** and error-prone; medical professionals need AI-assisted tools **for faster diagnosis, treatment planning, and reporting.**

- Built an **AI-powered medical chatbot** to process clinical reports (PDF, DOCX, JSON, CSV) — reducing manual review time by **50%.**
- Developed a **RAG pipeline** using FAISS vector database for context-aware, real-time responses from medical data, improving response accuracy by **23%.**
- Designed a **Flask + React web interface** with interactive chat, file upload, and automated PDF report generation — enabling clinicians to generate reports in **under 2 minutes.**
- Integrated advanced **NLP models** (Hugging Face Transformers, Cohere, LangChain) for **diagnosis and intelligent medical recommendations** — increasing **recommendation relevance by 20%.**
- Containerized the chatbot with **Docker, enabling scalable real-time querying** (e.g., clinicians upload reports and immediately ask “**What kind of diagnosis do you have?**”).

Certifications / Training

- **AWS Machine Learning Certified** [Link](#)
- **Professional Program on Data Science and AI (360DigiTMG)** [Link](#)
- **Microsoft Power BI** [Link](#)

Education

- **B.Tech in Computer Science** **2018 – 2022**
Savitribai Phule Pune University, Pune
- **12th Standard (HSC)** **2017 – 2018**
Shree Venkatesh Mahavidyalaya, Walandi